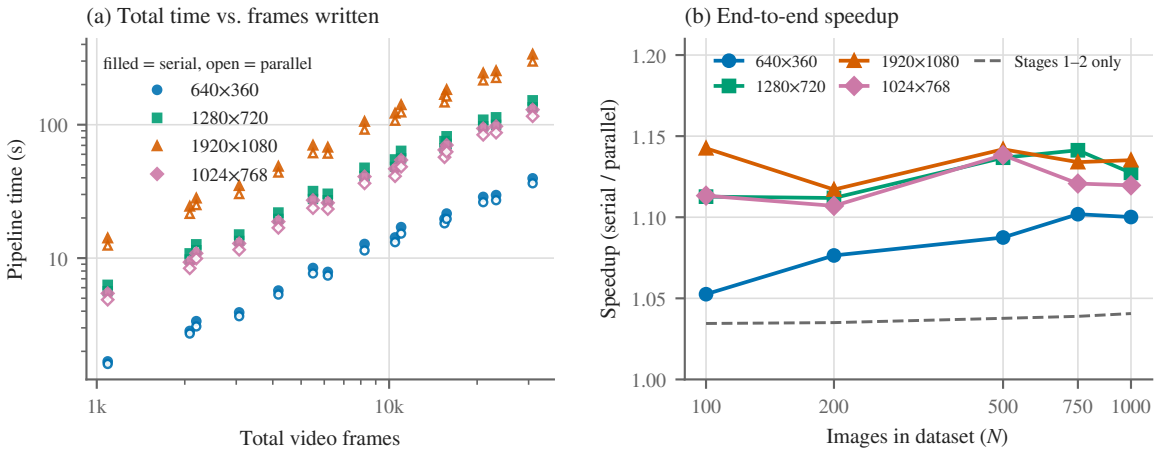
(a) Total time vs. frames written
filled = serial, open = parallel
640×360
1280×720
1920×1080
1024×768
Pipeline time (s)
100
10
1k
10k
Total video frames
(b) End-to-end speedup
640×360
1920×1080
Stages 1–2 only
1280×720
1024×768
Speedup (serial / parallel)
1.20
1.15
1.10
1.05
1.00
100
200
500
750
1000
Images in dataset (N)